S2.2 The covalent model

Strength of intermolecular forces

Learning outcomes

By the end of this section you should be able to:

- Compare the relative strengths of the types of intermolecular forces.
- Discuss the influence of molecular size and polarity on the strength of London (dispersion) forces.
- Explain the physical properties of covalent substances to include volatility, electrical conductivity and solubility in terms of their structure.

In section S1.1.2a (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/states-of-matter-and-changes-of-state-id-43067/) you learned about kinetic molecular theory and how particles are arranged in solids, liquids and gases. You also learned that conversion from one phase of matter to another phase is possible if sufficient energy is added or removed (**Figure 1**). Therefore, boiling point can be used as a measure of the attraction from one molecule to another. It measures the energy required to overcome the attraction between molecules and as they are separated (remember that gaseous molecules are found much further apart than liquid molecules).

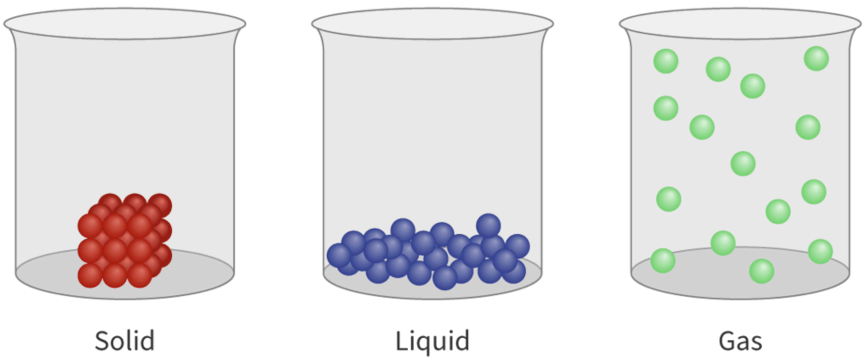

Figure 1. Phases of matter.

More information for figure 1

The image shows three transparent beakers, each demonstrating a different state of matter: solid, liquid, and gas. In the first beaker labeled 'Solid,' red spheres are tightly clustered at the bottom, representing densely packed molecules. In the second beaker labeled 'Liquid,' blue spheres are loosely packed at the bottom, indicating a more flexible arrangement than solids, allowing for some movement. In the third beaker labeled 'Gas,' green spheres are scattered throughout the beaker, symbolizing widely spaced molecules that freely move around inside the container. This representation illustrates the kinetic molecular theory where particles in solids are tightly packed, in liquids are moderately spread, and in gases are widely spaced apart.

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Remember that when a substance boils, it is the forces of attraction between molecules being overcome - it does not involve the breaking of covalent bonds between the atoms in the molecule.

Let's compare three different molecules: methane (CH_4), hydrochloric acid (HCl) and water (H_2O). **Table 1** shows that these three molecules have very different boiling points.

Table 1. Comparison of boiling points.

Molecule	Boiling point (°C)
Methane (CH_4)	−162
Hydrochloric acid (HCl)	−85
Water (H_2O)	100

If the molecules had a strong attraction to each other, would you expect the boiling point to be high or low?

A strong attraction indicates it would take a lot of energy to separate them. The fact that it takes very different amounts of energy for these three molecules to be separated suggests that they have different levels of attraction for each other. Why is this the case?

Factors affecting strength of force: type of force

As identified previously, all the intermolecular forces are not of equal strength. In **Figure 3** we can see the compounds formed when group 14 elements bond to hydrogen. While the bonds may be polar, due to the symmetry of the molecule, there is no net dipole moment. Therefore, these molecules will only experience London dispersion forces.

These forces seem to increase as you move down through group 14, we will discuss this further below.

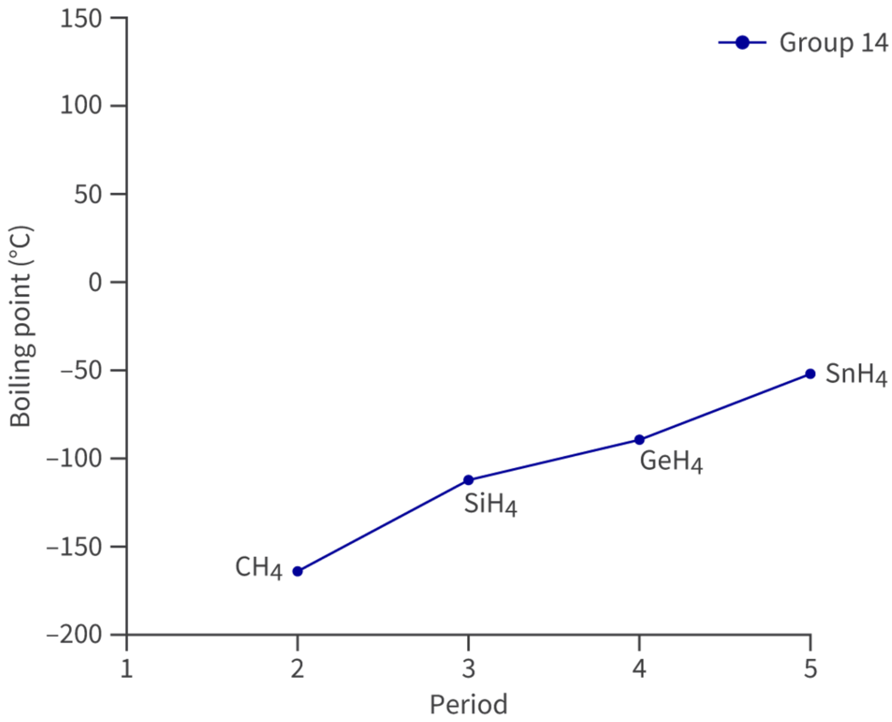

Figure 3. Boiling points of group 14 hydrides.

More information for figure 3

The image is a line graph representing the boiling points of Group 14 hydrides across different periods. The X-axis is labeled as 'Period' with integer values ranging from 1 to 5. The Y-axis is labeled as 'Boiling point (°C)' with values ranging from -200 to 150 °C at intervals of 50.

Data points for four compounds are plotted: 1. CH_4 at Period 1 with a boiling point of approximately -160°C. 2. SiH_4 at Period 3 with a boiling point of approximately -110°C. 3. GeH_4 at Period 4 with a boiling point of approximately -90°C. 4. SnH_4 at Period 5 with a boiling point of approximately -50°C.

The graph shows a clear upward trend, indicating that the boiling points increase as you go down the group, starting from CH_4 and ending at SnH_4.

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Compare this to some compounds formed when hydrogen bonds to elements from groups 15 to 17. As you can see in **Figure 4**, all the compounds in periods 3–5 for groups 15 to 17 have slightly greater boiling points than those in group 14. What intermolecular force is found between these molecules?

These compounds will experience both London dispersion **and** dipole–dipole forces. These dipole–dipole forces are stronger than London dispersion forces, which causes them to have a higher boiling point.

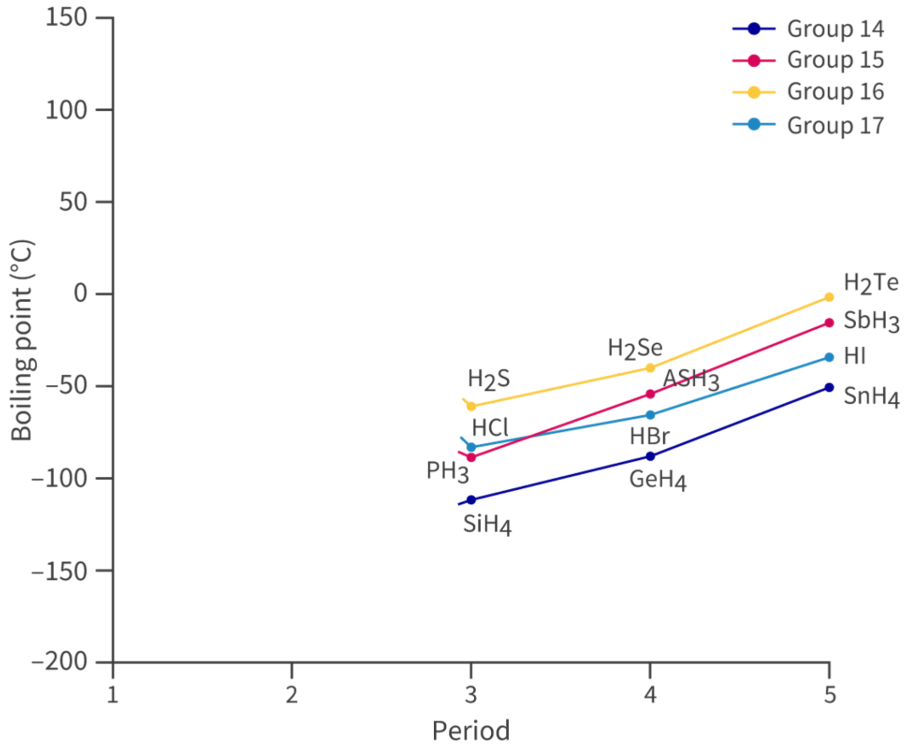

Figure 4. Boiling points of hydrogen and the compounds it forms with groups 14–17.

More information for figure 4

The graph displays the boiling points of hydrogen compounds with groups 14 to 17 elements across different periods indicated on the X-axis, ranging from periods 1 to 5. The Y-axis represents the boiling points in degrees Celsius, ranging from -200°C to 150°C.

The compounds are grouped and color-coded: - Group 14: SiH4, GeH4, SnH4 - Group 15: PH3, AsH3, SbH3 - Group 16: H2S, H2Se, H2Te - Group 17: HCl, HBr, HI

Key trends include a general increase in boiling points within each group as the period number increases.

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Figure 5 has been altered to include the compounds formed when period 2 elements from groups 14–17 bond to hydrogen. What do you notice about the graph in **Figure 5**? Can you identify the intermolecular force which occurs between these molecules?

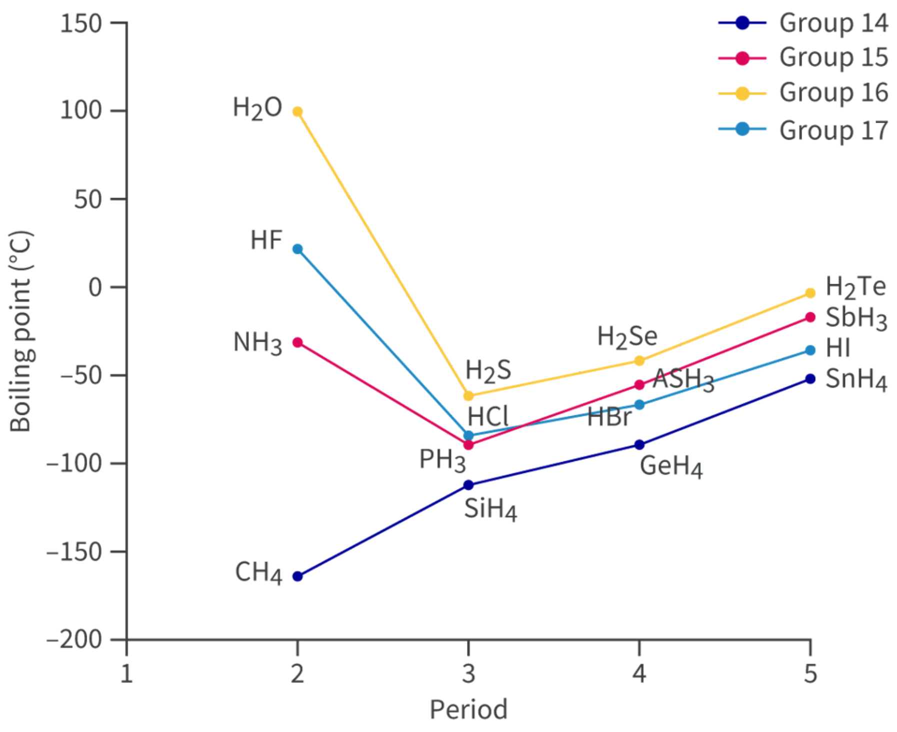

Figure 5. Boiling points of hydrogen and the compounds it forms with groups 14–17.

More information for figure 5

The image is a line graph illustrating the boiling points of hydrogen compounds formed with elements from groups 14 to 17 across periods 1 to 5. The x-axis represents the period number, ranging from 1 to 5, while the y-axis represents the boiling point in degrees Celsius, ranging from -200°C to 150°C. Four colored lines correspond to the groups: blue for Group 14, red for Group 15, yellow for Group 16, and cyan for Group 17, with each line indicating the trend of boiling points within that group across different periods.

Key data points include: - Group 14: Boiling points start from -165°C (CH4), rising slightly to -120°C (SiH4) and continuing to increase through GeH4. - Group 15: Starts at -33°C (NH3), dips to -90°C (PH3), and then gradually rises. - Group 16: Significant peak at 100°C (H2O), then decreasing (H2S, H2Se) before leveling. - Group 17: Begins at 20°C (HF), descending to nearly -85°C (HCl) and rising again (HBr, HI).

Overall, there's a distinct upward trend for compounds within individual groups, with notable peaks for hydrogen bonds, particularly in water and ammonia. The graph highlights hydrogen bonding effects, such as higher boiling points for H2O and NH3 due to stronger intermolecular forces.

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There is a clear discontinuity between the hydrogen compounds containing period 2 elements: the boiling points of these molecules are all much higher, indicating that the attraction between them requires a lot more energy to break.

The intermolecular force responsible for this is hydrogen bonding.

Hydrogen bonding is the strongest intermolecular force by far, and so the energy required to overcome it (the boiling point) is hugely increased.

Exercise 1

Click a question to answer

1. Which order is correct for molecules with comparable molar mass, when the following forces are arranged in order of **increasing** strength?

Check answers

Factors affecting the strength of forces: molecular size

The compounds formed between hydrogen and the group 14 elements all experience London dispersion forces but they do not have the same boiling point. This indicates that something other than the type of intermolecular force is impacting how strong the attraction is.

Study skills

Molecular size has a significant impact on the strength of the intermolecular force experienced. Therefore, any time you are comparing different compounds it is important that you specify if the molecular mass is staying the same or changing too.

As the molecular size of the compound increases, so does the strength of the force it increases. The greater the number of electrons in a molecule, the greater the probability of electron distribution asymmetry and the more susceptible the molecule is to developing an induced dipole from a charge nearby. As the strength of the attraction is greater, so is the energy required to overcome it.

Study skills

Although large molecules are more susceptible to an induced dipole due to differences in the size of their electron cloud, often this is described as an increase due to molecular mass — even though electrons have very little mass! Note that the supposed mass increase is related to the increase in protons, and therefore increase in electrons.

Properties of molecules

Electrical conductivity and melting and boiling points

Molecules are generally not electrically conductive. They do not have charged particles present, such as in ions in ionic compounds or delocalised electrons in carbon allotropes such as graphite and graphene.

They also do not have the same high melting and boiling points as ionic and metallic compounds. When an ionic or metallic compound are being melted or boiled, it requires a lot of energy to overcome the attractive forces within these materials. In the melting or boiling of a molecule, it is the comparatively weaker intermolecular forces being overcome, which requires significantly less energy.

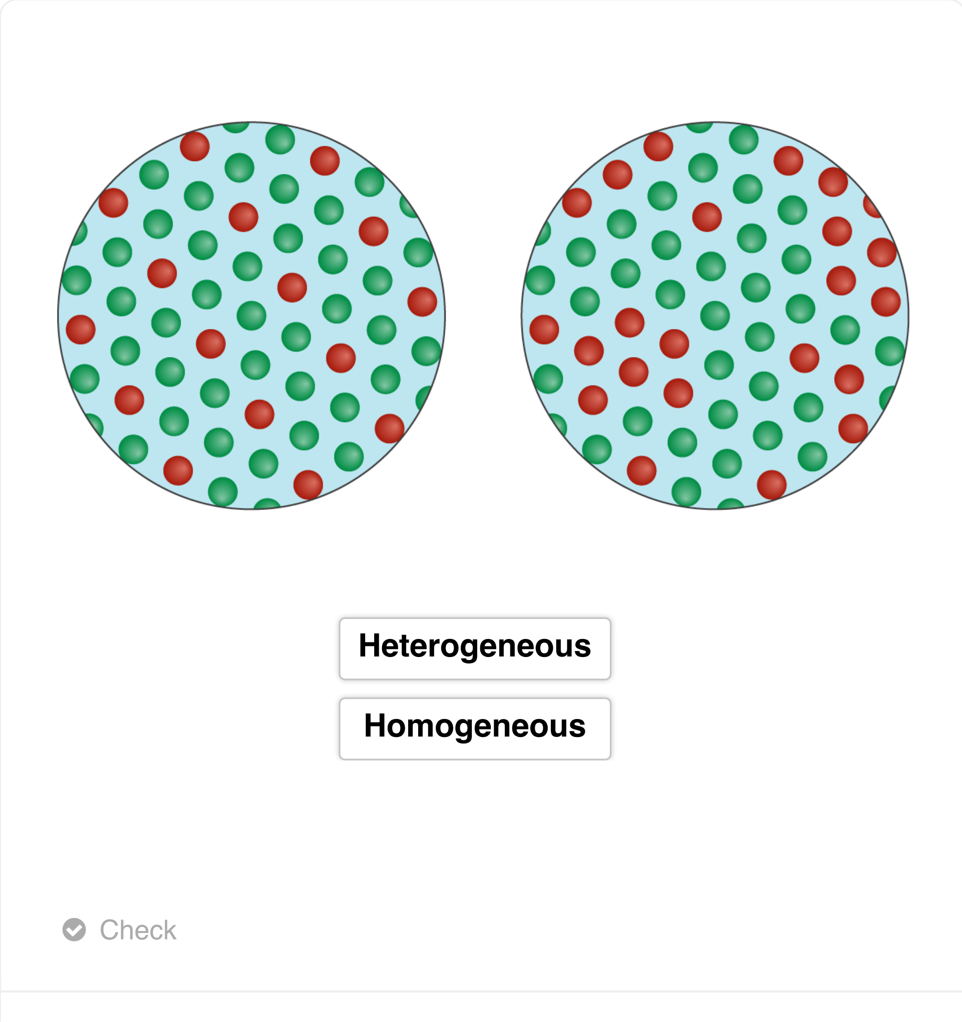
Volatility

The volatility of molecular elements and compounds depend on the type of intermolecular forces that exist between the molecules of the substance. Substances with stronger intermolecular forces tend to have low volatilities (and high boiling points) and those with weaker intermolecular forces tend to have high volatilities (and low boiling points). Let's consider the organic compound butane, C_4H_{10}, which has relatively weak London dispersion forces between its molecules. For this reason, butane is highly volatile with a low boiling point of -1 ^{O}C. Another organic compound ethanol, C_2H_5OH, has hydrogen bonding between its molecules. Hydrogen bonding is stronger than London dispersion forces, so ethanol is less volatile than butane and has a higher boiling point of 78 ^{O}C.

So far we have only considered molecular substances (those that form discrete molecules). But what about covalent network substances such as diamond and silicon dioxide? These types of compounds tend to have very low volatilities and very high boiling points because of the strong covalent bonds that exist between the atoms.

Solubility

Recall from section S1.1.1a (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/elements-compounds-and-mixtures-id-43065/) that there are two types of mixtures: homogeneous and heterogeneous mixtures. What is the difference between these two types? Click and drag labels to each of the images to identify them as homogeneous or heterogeneous mixtures.



Interactive 1. Mixtures can be Either Homogeneous or Heterogeneous.

More information for interactive 1

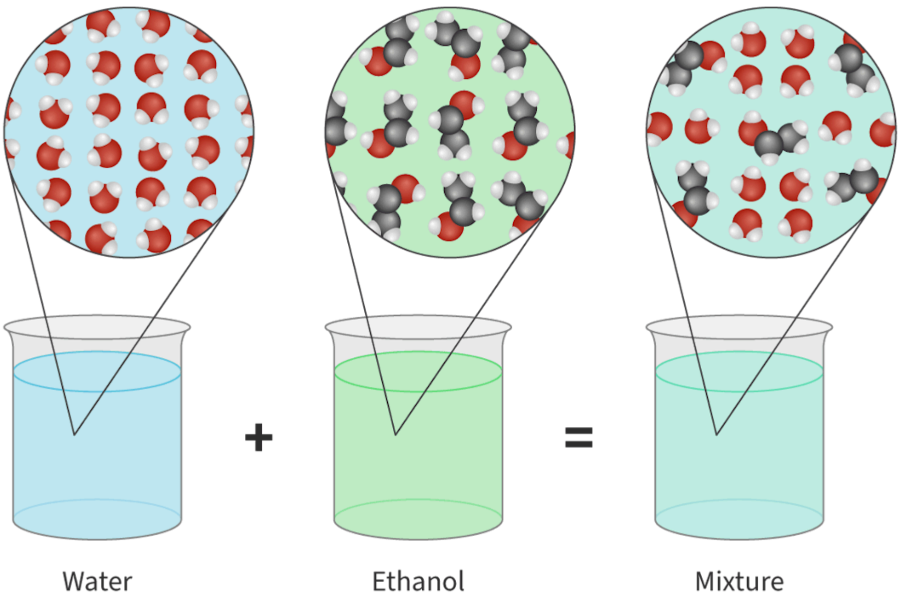
An interactive diagram features close-up views of two mixtures. The view on the left shows an orderly arrangement of red and green spheres. The view on the right shows a random arrangement of red and green spheres.
Users need to identify the type of mixture in each view by dragging and dropping the correct label from the options below:
Homogeneous
Heterogeneous
This interactivity helps users to differentiate between the homogeneous and heterogeneous mixtures based on particle distribution.
Read below for the solution:
Left view: Homogeneous
Right view: Heterogeneous

Molecules that are soluble in a substance will form a homogeneous mixture with it, with uniform properties throughout.

Certain molecules will be soluble in polar substances such as water while others will not. This can all be linked back to the type of intermolecular force that is found between the molecules. A general guideline is that 'like dissolves like': polar molecules will dissolve in polar solvents and non-polar molecules will dissolve in non-polar solvents forming homogeneous mixtures (**Figure 6**).

Iodine molecules, I_2, will dissolve in a non-polar solvent such as hexane due to iodine's ability to interact with the hexane via London dispersion forces. Iodine will form a homogenous mixture with hexane. Iodine would be insoluble in a polar solvent such as water as the London dispersion forces that iodine experiences are not sufficient to overcome the strong hydrogen bonds that water can form. It will form a heterogeneous mixture with the water.

Other polar substances, such as ethanol, can dissolve in water.



More information

The image consists of three beakers arranged sequentially from left to right. The first beaker is labeled 'Water' and is shown with a light blue liquid. Above it, a magnified view illustrates water molecules, depicted as clusters of red and white spheres, representing the oxygen and hydrogen atoms, respectively. The middle beaker is labeled 'Ethanol', containing a light green liquid, with a magnified view above showing ethanol molecules. These include additional gray spheres, representing the carbon atoms, along with the red and white spheres. The third beaker, labeled 'Mixture', contains a light blue-green liquid, indicating a combination of water and ethanol. Its magnification shows the intermixed water and ethanol molecules. A plus sign between the first two beakers and an equal sign between the last two indicates the mixing process of water and ethanol resulting in a solution.

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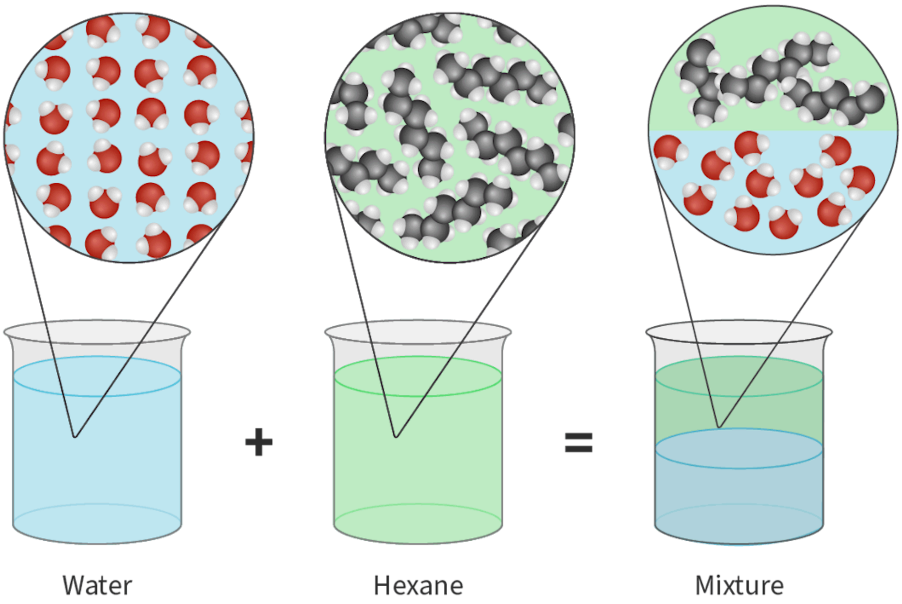

Figure 6. Like dissolves like: polar molecules will dissolve in polar solvents but non-polar molecules do not.

More information for figure 6

The image shows three beakers, each labeled: "Water," "Hexane," and "Mixture," respectively. Above each beaker is a magnified section depicting the molecular structure. The beaker labeled "Water" contains a blue liquid, and its magnified section shows water molecules with red and white atoms. The beaker labeled "Hexane" contains a green liquid, and its magnified section shows hexane molecules composed of black and white atoms. The third beaker labeled "Mixture" shows green and blue layers with magnified sections of both water and hexane molecules, indicating a separation of polar and non-polar molecules in solution.

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The solubility of some polar molecules will decrease as the size of the molecule increases, as the polar part of the molecule becomes relatively small compared to the size of the overall molecule. For example, hexanol will be much less soluble in water than ethanol.

Covalent network structures are generally insoluble in both polar and non-polar solvents as the energy required to overcome their covalent bonds is too high.

In the next activity, you will be given a number of cards featuring experimental data about different molecules. Analyse this data and use it, along with the knowledge you have gained from this subtopic, to answer the questions provided on each card.

Activity

- **IB learner profile attribute:** Thinker
- **Approaches to learning:** Thinking skills — Applying key ideas and facts in new contexts
- **Time required to complete activity:** 30 minutes
- **Activity type:** Individual or group activity

If working as an individual: You will be given cards, each one containing a 'prompt question' and experimental data relating to intermolecular forces. Analyse the data given and answer each prompt.

If working in a group: Each person in your group is to be given a card with a 'prompt question' and experimental data relating to intermolecular forces. Complete one card each and then share your findings with the rest of your group.

Prompt questions
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